COMPUTING HOMOTOPY GROUPS OF MADSEN-TILLMANN SPECTRA

BRIAN CECO

ABSTRACT. The unoriented Madsen-Tillmann spectrum $MTO(2)$ has been extensively studied by geometric topologists. The connected component of its loop space is $B\mathcal{N}_\infty^+$, the (Quillen plus construction of the) classifying space of the stable non-orientable mapping class group. Away from the prime 2, the homotopy type of $B\mathcal{N}_\infty^+$ is well-known (it coincides with the connected component of the free infinite loop space of $\mathbf{HP}^\infty$). There is ongoing work to understand the situation at the prime 2. We utilize Rognes' computations of differentials in the Adams spectral sequence for $MTSO(2)$ along with the splitting theorems of Kashiwabara and Zare to compute the 2-primary homotopy groups of $MTO(2)$, and consequently the stable homotopy groups of BN_∞, up to degree 15. We hope to use these computations in the future to extend the range of known computations of the rank of the $\mathbf{Z}/2$-homology groups of BN_∞^+ in low degrees.

CONTENTS

1. INTRODUCTION

In [GMTW09], Galatius, Madsen, Tillmann, and Weiss defined for each nonnegative integer $d \geq 0$, a spectrum $MT\theta(d)$, whose infinite loop space deloops to the classifying space $B\mathsf{Cob}(d)_\theta$ of a particular topological category. The category $\mathsf{Cob}(d)_\theta$ parametrizes cobordisms between $(d-1)$-dimensional closed smooth manifolds with θ-structures for a given tangential structure $\theta : B \to BO(d)$. The homotopy invariants of these spectra, called *Madsen-Tillmann spectra*, contain information about diffeomorphism groups of manifolds [GRW10], as well as applications to physics via topological quantum field theories [FH21]. These applications have sparked great interest in using techniques from stable homotopy theory to better understand and classify these spectra.

One of the most effective means of studying Madsen-Tillman spectra came in the work of Kashiwabara and Zare [KZ18], which made use of the transfer maps of [BG74] and [MMM86] to give splittings of several classes of Madsen-Tillmann spectra localized at various primes. We focus our attention on the unoriented Madsen-Tillmann spectra $MTO(n)$. Kashiwabara and Zare determined the homotopy type of $MTO(n)$ away from the prime 2 .

Theorem 1.0.1. *Given $n \geq 0$, we have a homotopy equivalence*

$$MTO(n)\left[\frac{1}{2}\right] \simeq \begin{cases} \Sigma^\infty_+ BO(n)\left[\frac{1}{2}\right], & n \text{ even}, \\ *, & n \text{ odd}. \end{cases}$$

In the case $n = 2$, in a follow-up paper [KZ15], Kashiwabara and Zare were able to determine the homotopy type of $MTO(2)$ completed at the prime 2, hence effectively giving a complete determination of its homotopy type.

Theorem 1.0.2 (Corollary 4.4, [KZ15])**.** *We have a homotopy equivalence of spectra*

$$MTO(2)^{\wedge}_{2} \simeq \Sigma^{\infty}_{+} BSO(3)^{\wedge}_{2} \vee \Sigma^{-2} D(2).$$

We use the above splitting theorem to extend the range of known computations of the 2-primary homotopy groups of $MTO(2)$ (modulo some extension problems) through dimensions 15. The results are summarized by the table 4.4. Our strategy follows the standard tried-and-true method for carrying out computations within the stable homotopy category—we make use of the Adams spectral sequence. As a warm-up to demonstrate the techniques we employ for our desired computation, we also determine all the differentials in the Adams spectral sequence for $MTO(1)$ up through topological degree $t - s \leq 38$.

1.1. **Acknowledgements.** I would like to thank my thesis advisor Professor Sander Kupers for the indispensable guidance and support he provided throughout the writing of this thesis, without which this work would not be possible.

2. Madsen-Tillmann spectra and homological stability

We begin by giving a brief overview of the historical motivation behind the construction of Madsen-Tillmann spectra. Our reason for doing this is two fold—partly to justify our interest in the homotopy groups of $MTO(2)$, as well as to recall some key definitions and results to be used throughout the paper.

2.1. **The oriented case.** Let $\Sigma_{g,b}$ denote a connected compact oriented surface of genus g with b many boundary components.

Definition 2.1.1. The *mapping class group* $\Gamma_{g,b}$ is the group of isotopy classes of oriented diffeomorphisms of $\Sigma_{g,b}$ which fix the boundary pointwise:

$$\Gamma_{g,b} \coloneqq \pi_0(\mathrm{Diff}^{+}_{\partial}(\Sigma_{g,b})).$$

We write Γ_g for $\Gamma_{g,0}$.

There are various maps between the oriented surfaces $\Sigma_{g,b}$ for different values of (g, b), which induce associated homomorphisms between their mapping class groups. Namely, if $b > 0$, we have a homomorphism $\alpha\colon \Gamma_{g,b} \to \Gamma_{g,b+1}$ induced by gluing a pair of pants along a single boundary component and extending diffeomorphisms in the obvious way. Similarly, if $b > 1$, we have a homomorphism $\delta\colon \Gamma_{g,b} \to \Gamma_{g+1,b-1}$ induced by gluing a pair of pants along two boundary components.

Harer [Har85] showed that α and δ induce isomorphisms on integral homology groups in dimensions $k \leq (g-3)/3$. Shortly after, Ivanov [Iva90] extended these computations, improving Harer's bounds to $(g-1)/2$ and $(g-2)/2$ respectively. Given $k \geq 0$, we refer to those values of $g \geq 0$ for which we know the above maps induce isomorphisms on $H_k(-; \mathbf{Z})$ (i.e., $g \geq 2k + 2$ by Ivanov's work) as the *stable range*. Thus $H_k(\Gamma_{g,b}; \mathbf{Z})$ is independent of b and g for g within the stable range. These are called the *stable homology groups* of the mapping class groups.

Definition 2.1.2. The *stable mapping class group* $\Gamma_{\infty,b}$ is defined as the colimit

$$\Gamma_{\infty,b} \coloneqq \operatorname*{colim}_{n\to\infty} \Gamma_{n,b},$$

where the structure homomorphisms $\Gamma_{n,b} \to \Gamma_{n+1,b}$ are given by the composite

$$\Gamma_{n,b} \xrightarrow{\alpha} \Gamma_{n,b+1} \xrightarrow{\delta} \Gamma_{n+1,b}.$$

Again, for $b = 0$, we suppress the subscript, writing Γ_∞.

The homology of the stable mapping class group Γ_∞ is the stable homology of the mapping class groups, i.e. the stability theorem gives isomorphisms

$$H_k(\Gamma_\infty) \cong H_k(\Gamma_{g,b})$$

for $g \geq 2k + 1$ if $b \geq 1$ and $g \geq 2k$.

The group Γ_g is perfect (i.e., its abelianization G^{ab} is trivial) for $g \geq 3$, and hence Γ_∞ is also perfect (as $(-)^{\mathrm{ab}}$ is a left adjoint and hence commutes with colimits). In particular, the classifying space $B\Gamma_\infty$ is a connected aspherical CW-complex with $\pi_1(B\Gamma_\infty) \cong \pi_0(\Gamma_\infty) = \Gamma_\infty$. Applying the Quillen plus construction to the classifying space of the infinite mapping class group yields a simply connected space $B\Gamma_\infty^+$.

Importantly, the effect the Quillen plus construction has on the higher homotopy groups of the initial space is in general not clear. In the case of Γ_∞^+, at the expense of killing off π_1, we have created infinitely many nontrivial homotopy groups. An upside to passing to the plus construction is that for $b \geq 1$, the homology equivalence $B\Gamma_{\infty,b} \to B\Gamma_\infty$ induced by the composite β^b induces homotopy equivalences $B\Gamma_\infty^+ \simeq B\Gamma_{\infty,b}^+$ by Whitehead's theorem, or alternatively, via the universal property of the plus construction.

In [Til97], Tillmann determined $\mathbf{Z} \times B\Gamma_\infty^+$ has the homotopy type of $\Omega B\mathcal{C}$, where $\mathcal{C}$ was a particular symmetric monoidal simplicial cobordism category of oriented surfaces. Segal's infinite loop space machinery implies that the classifying space of any topological symmetric monoidal category $\mathcal{M}$ is an $\mathbf{E}_\infty$-space. In particular, the corresponding loop space $\Omega B\mathcal{M}$ is grouplike, and so $\mathbf{Z} \times B\Gamma_\infty^+$ has the homotopy type of an infinite loop space. One is then naturally led to ask if there exist any interesting spectra whose connective cover realize the spectrum given by delooping $\mathbf{Z} \times B\Gamma_\infty^+$. This is precisely where Madsen-Tillmann spectra enter the picture.

Definition 2.1.3. Given a natural number $n \geq 0$, fix an n-dimensional tangential structure $B_n \to BO(n)$. Let $\gamma_n^B \to B_n$ denote the universal n-plane bundle with B_n-structure. The *Madsen-Tillmann spectrum MTB_n* is defined as the Thom spectrum

$$MTB_n \coloneqq B_n^{-\gamma_n^B}.$$

For our purposes, we will only be interested in the tangential structures $B_n = BK(n)$, where $K(n)$ denotes one of $O(n)$ or $SO(n)$, with the corresponding obvious fibration $BK(n) \to BO(n)$. The associated Madsen-Tillmann spectra are denoted by $MTO(n)$ and $MTSO(n)$ respectively. In the landmark paper resolving the Mumford conjecture [MW07], Madsen and Weiss established a homotopy equivalence

$$\alpha_\infty^{SO} \colon \mathbf{Z} \times B\Gamma_\infty^+ \xrightarrow{\simeq} \Omega^\infty MTSO(2).$$

Remark 2.1.4. There is an analogous mapping class group construction for non-orientable surfaces. Let $N_{g,b}$ denote a connected compact non-orientable surface of (non-orientable) genus g[1] and b boundary components. By the classification of compact connected surfaces, $N_{g,b}$ has the diffeomorphism type of the g-fold connected sum $(\mathbf{RP}^2)^{\# g}$ with b disjoint discs removed. We define the *non-orientable mapping class group $\mathcal{N}_{g,b}$* of $N_{g,b}$ analogously to the oriented case 2.1.1, though of course we drop the requirement of the diffeomorphisms being orientation-preserving. Natalie Wahl [Wah08] established an analogue of Harer's homological stability theorem for the non-orientable mapping class groups. Although $\mathcal{N}_\infty$ is not a perfect group as in the oriented case, its maximal normal subgroup has index 2, and as before we can apply the Quillen plus construction to yield a space $B\mathcal{N}_\infty^+$. This space is also an infinite loop space, corresponding to the connective cover of the Madsen-Tillmann spectrum $MTO(2)$.

2.2. A cofiber sequence of Madsen-Tillmann spectra.

Notation 2.2.1. Throughout, $K(n)$ denotes either $O(n)$ or $SO(n)$

Let $M \to E \xrightarrow{f} B$ be a smooth fiber bundle with fiber M^n a closed n-manifold with $K(n)$-structure. In particular, f is a surjective submersion, hence we have a short exact sequence of vector bundles on E

$$0 \longrightarrow T_f E \longrightarrow TE \longrightarrow f^*TB \longrightarrow 0 \tag{2.2}$$

where the kernel $T_f E$ is called the *vertical or (fiberwise) tangent bundle* of f. Note $T_f E$ naturally inherits a $K(n)$-structure, hence it is classified by a map $\psi \colon E \to BK(n)$, unique up to homotopy. This yields a map at the level of Thom spectra:

$$\kappa_E \colon E^{-T_f E} \to MTK(n).$$

Fix a smooth embedding $i \colon E \hookrightarrow \mathbf{R}^n$. From this, we define a fiberwise inclusion $j = (f, i) \colon E \to \underline{n}_B$ with normal bundle ν_j. We have the following short exact sequence of vector bundles on E:

$$0 \longrightarrow TE \longrightarrow f^*TB \oplus \underline{n}_E \longrightarrow \nu_j \longrightarrow 0. \tag{2.3}$$

[1]The non-orientable genus $k(N)$ for a non-orientable surface N is defined via the Euler characteristic as $k(N) \coloneqq 2 - \chi(N)$.

Combining the short exact sequences 2.2 and 2.3 above yields a new short exact sequence

$$0 \longrightarrow T_f E \longrightarrow \boldsymbol{n}_E \longrightarrow \nu_j \longrightarrow 0,$$

hence we get an identification of virtual vector bundles $-T_f E = \nu_j - \boldsymbol{n}_E$. The Pontryagin-Thom collapse applied to j yields a map of Thom spectra

$$\Sigma^n f^! \colon \Sigma^{\infty+n}_+ B \to \Sigma^n E^{-T_f E}.$$

Definition 2.2.4. We define the *Madsen-Tillmann-Weiss map* (or *MTW map*) α_E of the bundle f as the composite

$$\alpha_E \colon \Sigma^\infty_+ B \xrightarrow{f^!} E^{-T_f E} \xrightarrow{\kappa_E} MTK(n)\ .$$

Example 2.2.5. The sphere bundle Bi associated to the canonical embedding $i\colon K(n) \hookrightarrow K(n+1)$ induces an MTW-map $\Sigma^\infty_+ BK(n+1) \to MTK(n)$ which we denote by $\alpha_{K(n)}$.

The following is our main computational tool for computing homotopy groups of Madsen-Tillmann spectra.

Proposition 2.2.6 (Proposition 3.1 of [GMTW09])**.** *We have a cofiber sequence*

$$MTK(n+1) \xrightarrow{\beta} \Sigma^\infty_+ BK(n+1) \xrightarrow{\alpha_{K(n)}} MTK(n) \xrightarrow{\eta} \Sigma MTK(n+1)$$

which we call the Madsen-Tillmann Weiss (MTW) cofiber sequence associated to $MTK(n+1)$.

The map β is induced by the inclusion of virtual vector bundles $-\gamma^K_{n+1} \to \mathbf{n}$ (upon passing to the finite sub-skeleta of the base $BK(n+1))$[2], and the map η is the virtual analog of the n^{th} structure map of the Thom spectrum MK.

3. Splitting Madsen-Tillmann spectra

The main insight we will leverage to compute the homotopy groups of $MTO(2)$ is the splitting result 1.0.2 of Kashiwabara and Zare. The summand $D(2)$ is part of a sequence of spectra $D(n)$, $n \geq 0$, introduced by Mitchell and Priddy in [MP83] in terms of symmetric products of spectra. We briefly motivate and recount their construction.

Convention 3.0.1. From this point on, we assume all spectra are implicitly completed at the prime 2, and all ordinary cohomology is taken with $\mathbf{Z}/2$-coefficients.

We begin by defining a number of modules over the mod 2 Steenrod algebra $\mathcal{A}$, each one realized as the cohomology of a spectrum arising from constructions involving the symmetric products of the sphere spectrum.

Definition 3.0.2. Let $I = (i_1, \dots, i_n)$ be a sequence of nonnegative integers $i_k \geq 0$. The *length* $l(I)$ of I is the number of nonzero terms in the sequence.

Definition 3.0.3. Fix an integer $n \geq 0$.

1. We let D_n denote the vector subspace of $\mathcal{A}$ spanned by monomials Sq^I of length $\ell(I) \leq n$.
2. We let G_n denote the vector subspace of $\mathcal{A}$ spanned by admissible monomials Sq^I of length $\ell(I) \geq n$.

It is immediately clear from the definition along with the Adem relations that G_n is an $\mathcal{A}$-module. We need some further justification that the $\mathbf{Z}/2$-vector space D_n is indeed an $\mathcal{A}$-module, i.e., closed under left multiplication by Steenrod squares.

Firstly, note that given a Steenrod square Sq^I of length $l(I) = n$, applying the Adem relations expresses Sq^I as unique linear combination of admissible monomials, each of length *at most* n, hence D_n has a basis consisting of *admissible* monomials of length at most n.

Given $n \geq 0$, the module G_n is in fact a left ideal of $\mathcal{A}$. Furthermore, one readily sees that the quotient $\mathcal{A}$-module $\mathcal{A}/G_{n+1}$ has a vector space basis given by admissible monomials of length at most n, i.e., $\mathcal{A}/G_{n+1}$ realizes D_n as an $\mathcal{A}$-module.

We also have inclusions of left ideals $G_{n+1} \subset G_n$ for every $n \geq 0$. The corresponding subquotient

$$M_n \coloneqq G_n / G_{n+1}$$

[2] See Remark 4.1.2

of $\mathcal{A}$ canonically embeds into $\mathcal{A}$ as the linear subspace spanned by admissible sequences Sq^I of length $l(I) = n$.

Recall, for an integer $n \geq 0$ and a spectrum X, we can define the n^{th} *symmetric product* $\mathrm{SP}^n(X)$ of X. This construction is functorial in X (in the stable homotopy category), and furthermore, we have comparison maps $i_n\colon \mathrm{SP}^n(X) \to \mathrm{SP}^{n+1}(X)$. Passing to the homotopy colimit allows us to define an *infinite symmetric product* $\mathrm{SP}^\infty(X)$. In the case $X = \mathbf{S}^0$, we have $\mathrm{SP}^\infty(\mathbf{S}^0) \simeq H\mathbf{Z}$. Nakaoka determined that the filtration

$$\mathbf{S}^0 \longrightarrow \mathrm{SP}^2(\mathbf{S}^0) \longrightarrow \mathrm{SP}^{2^2}(\mathbf{S}^0) \longrightarrow \ldots \longrightarrow \mathrm{SP}^{2^n}(\mathbf{S}^0) \longrightarrow \ldots \longrightarrow H\mathbf{Z}$$

realizes the *length filtration* on $H^*(H\mathbf{Z}) \cong \mathcal{A}/\mathcal{A}Sq^1$, in the following sense. First, we need some notation.

Notation 3.0.4. Given an $\mathcal{A}$-module M, denote by $\widetilde{M}$ the corresponding $\mathcal{A}/Sq^1\mathcal{A}$-module induced by the extension of scalars along the ring homomorphism $\mathcal{A} \to \mathcal{A}/Sq^1\mathcal{A}$.

Theorem 3.0.5 ([Nak58])**.** *Fix an integer $n \geq 0$.*

1. *If $n+1 \neq 2^k$ for some $k \geq 0$, the map $i_n : \mathrm{SP}^n(\mathbf{S}^0) \to \mathrm{SP}^{n+1}(\mathbf{S}^0)$ is a homotopy equivalence.*
2. *We have an isomorphism of $\mathcal{A}$-modules*

$$H^*(\mathrm{SP}^{2^n}(\mathbf{S}^0)) \cong \widetilde{D}_n$$

and furthermore, the fiber sequence associated to the induced map $\mathrm{SP}^n(\mathbf{S}^0) \to H\mathbf{Z}$ realizes the short exact sequence of $\mathcal{A}$-modules

$$0 \longrightarrow \widetilde{G}_{n+1} \longrightarrow \mathcal{A}/Sq^1\mathcal{A} \longrightarrow \widetilde{D}_n \longrightarrow 0\ .$$

Given a space X, the l-fold diagonal $\Delta_l\colon X \to X^l$ induces by naturality a map of symmetric products

$$\Delta_l'\colon\ \mathrm{SP}^n(X) \to \mathrm{SP}^n(X^l).$$

The natural map $(X^l)^n \simeq X^{ln} \to \mathrm{SP}^{ln}(X)$ factors through the Σ_n-orbits of the source, hence we can consider the composite map

$$\Delta_l\colon \mathrm{SP}^n(X) \xrightarrow{\Delta_l'} \mathrm{SP}^n(X^l) \longrightarrow \mathrm{SP}^{ln}(X).$$

This space-level construction induces a map of prespectra, which by the functoriality of fibrant replacement, yields a natural transformation $\mathrm{SP}^n(-) \to \mathrm{SP}^{ln}(-)$ of endofunctors on $\mathrm{Ho}(\mathsf{Sp})$.

Given $n \geq 1$, we define the spectrum $D(n)$ to be the cofiber of the map

$$\Delta_2\colon\ \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) \to \mathrm{SP}^{2^n}(\mathbf{S}^0).$$

By convention $\mathrm{SP}^{2^{-1}}(\mathbf{S}^0)$ has the homotopy type of a point, so that $D(0) = \mathrm{SP}^1(\mathbf{S}^0) = \mathbf{S}^0$.

We have a cofiber sequence of filtered spectra

$$\begin{tikzcd}
* \arrow[r] \arrow[d, "\Delta_p"'] & \mathbf{S}^0 \arrow[r] \arrow[d, "\Delta_p"'] & \mathrm{SP}^2(\mathbf{S}^0) \arrow[r] \arrow[d, "\Delta_p"'] & \ldots \arrow[r] & H\mathbf{Z} \arrow[d, "p"] \\
\mathbf{S}^0 \arrow[r] \arrow[d] & \mathrm{SP}^2(\mathbf{S}^0) \arrow[r] \arrow[d] & \mathrm{SP}^{2^2}(\mathbf{S}^0) \arrow[r] \arrow[d] & \ldots \arrow[r] & H\mathbf{Z} \arrow[d] \\
\mathbf{S}^0 \arrow[r] & D(1) \arrow[r] & D(2) \arrow[r] & \ldots \arrow[r] & H\mathbf{Z}/p
\end{tikzcd}.$$

Mitchell and Priddy go on to prove a realization theorem for the above filtration of $H\mathbf{Z}/p$ analogous to Nakaoka's theorem for $H\mathbf{Z}$.

Theorem 3.0.6. *We have an isomorphism of $\mathcal{A}$-modules*

$$H^*(D(n); \mathbf{Z}/2) \cong \mathcal{A}/G_{n+1}$$

and furthermore, the map $D(n) \to H\mathbf{Z}/2$ realizes the projection map $\mathcal{A} \to \mathcal{A}/G_{n+1}$ on $\mathbf{Z}/2$-cohomology.

Finally, given $n \geq 0$, we define the spectrum $L(n)$ as the n-fold desuspension of the cofiber of the map $j_n\colon\ \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) \to \mathrm{SP}^{2^n}(\mathbf{S}^0)$ corresponding to the inclusion $(\mathbf{S}^0)^{2^{n-1}} \to (\mathbf{S}^0)^{2^n}$ of the first 2^{n-1}-factors. The situation is summarized by the following commutative diagram of spectra, where the rows and the columns are cofiber sequences:

$$\begin{tikzcd} \mathrm{SP}^{2^{n-2}}(\mathbf{S}^0) \arrow[r, "\Delta_p"] \arrow[d, "j_n"'] & \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) \arrow[r] \arrow[d, "j_{n+1}"'] & D(n-1) \arrow[d] \\ \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) \arrow[r, "\Delta_p"] \arrow[d] & \mathrm{SP}^{2^n}(\mathbf{S}^0) \arrow[r] \arrow[d] & D(n) \arrow[d] \\ \Sigma^{n-1}L(n-1) \arrow[r] & \Sigma^n L(n) \arrow[r] & \Sigma^n M(n) \end{tikzcd} \tag{3.7}$$

For $n = 1$, Mitchell and Priddy showed that the above cofiber sequences take on a familiar form.

Proposition 3.0.8 (Theorem 3.2 of [MP84])**.** *We have homotopy equivalences*

$$D(1) \simeq \Sigma MTO(1), \quad L(1) \simeq \Sigma^\infty \mathbf{RP}^\infty, \quad M(2) \simeq \Sigma^\infty_+ \mathbf{RP}^\infty$$

and the rightmost column in the commutative square 3.7 is the MTW cofiber sequence associated to $MTO(1)$.

4. Computing the homotopy groups of $MTO(2)$

4.1. **A word on the Adams spectral sequence.** The Adams spectral sequence is among the most powerful computational tools in stable homotopy theory. Informally, it is a way of extracting homotopical information (namely, about the graded abelian group of homotopy classes of maps between two spectra) via (co)homological inputs. For the computations we are interested in, we will make use of the classical mod 2 (cohomological) Adams spectral sequence for various spectra X (implicitly 2-complete), which takes the form

$$E_2^{s,t} = \mathrm{Ext}_{\mathcal{A}}(H^*(X), \mathbf{Z}/2) \Rightarrow \pi_{t-s}(X) \tag{4.1}$$

In order to compute the E_2-page of the Adams spectral sequence of X, one constructs a *minimal resolution* of $H^*(X)$ as a module over $\mathcal{A}$. This is a projective resolution $P_\bullet \to H^*(X)$ which, upon applying the internal-hom in the category of graded $\mathcal{A}$-modules, yields a cochain complex

$$\underline{\mathrm{Hom}}_{\mathcal{A}}(P_0, \mathbf{Z}/2) \xrightarrow{\partial_1^*} \underline{\mathrm{Hom}}_{\mathcal{A}}(P_1, k) \xrightarrow{\partial_2^*} \dots$$

such that $\partial_s^* = 0$ for all $s \geq 0$. In particular, the cohomology of this complex (i.e., the $\mathrm{Ext}_{\mathcal{A}}(-, \mathbf{Z}/2)$-groups of $H^*(X)$) can be read off without any reference to subquotients. As $\mathcal{A}$ is a connected Hopf algebra, every $\mathcal{A}$-module admits a minimal resolution, and furthermore, if $H^*(X)$ is bounded below (e.g., if X is eventually connective), we can find minimal resolutions by free $\mathcal{A}$-modules.

While it is relatively straightforward to compute minimal resolutions by hand in low degrees, we make use of Bruner's wonderful Ext software [Bru22] to compute minimal resolutions and produce the Adams E_2-pages depicted in the Appendix A.

Remark 4.1.2 (Cohomology of Madsen-Tillmann spectra)**.** Denote by $\mathrm{Th}(\gamma_n^K)$ the Thom space of the universal $K(n)$-bundle γ_n^K. Let $U \in \widetilde{H}^n(\mathrm{Th}(\gamma_n^K))$ denote the Thom class of γ_n^K, which is guaranteed to exist as we are asking for $\mathbf{Z}/2$-orientability of our bundles. The Thom spectrum $MTK(n)$ admits a cohomology class $\overline{U} \in H^{-n}(MTK(n))$, called a *virtual Thom class*, such that $H^*(MTK(n))$ is the free $H^*(BK(n))$-module on $\overline{U}$. Note that while the definition of virtual Thom classes for vector bundles $V \to X$ over a compact base space X is straightforward, as we can represent $-V$ as the formal difference of a real vector bundle by a trivial bundle, the general case require a bit more care. Details can be found in section 5 of [GMTW09]. Importantly, our computations are greatly facilitated by the fact that the $\mathcal{A}$-module structure of $H^*(BO(n))$ and $H^*(BSO(n))$ is completely known in terms of the Wu formula for Stiefel-Whitney classes. A good reference for this material is [MS74].

In particular, the above remark implies the MTW cofiber sequence 2.2.6 induces (up to a suspension) a short exact sequence of $\mathcal{A}$-modules.

Proposition 4.1.3. *Let $K(n)$ denote either $O(n)$ or $SO(n)$ for a fixed integer $n \geq 0$. The suspended MTW cofiber sequence*

$$MTK(n) \xrightarrow{\eta} \Sigma MTK(n+1) \xrightarrow{-\Sigma\beta} \Sigma^{\infty+1}_+ BK(n+1) \xrightarrow{-\Sigma\alpha_{K(n)}} \Sigma MTK(n)$$

induces a short exact sequence on cohomology groups.

Proof. Writing $\beta^* := (-\Sigma\beta)$, and $\partial^* = (-\Sigma\alpha_{K(n)})$, the MTW cofiber sequence induces a long exact sequence of cohomology groups

$$\ldots \longrightarrow H^k(\Sigma BK(n)) \xrightarrow{\beta^*} H^k(\Sigma MTK(n+1)) \xrightarrow{\eta^*} H^k(MTK(n)) \xrightarrow{\partial^*} H^{k+1}(\Sigma BK(n+1)) \longrightarrow \ldots .$$

It suffices to prove the boundary map $\partial^* \colon H^*(MTK(n)) \to H^{*+1}(BK(n+1))$ is zero. This can be seen in two ways. To have no ambiguity on the existence of a first Stiefel-Whitney, let us assume $K(n) = O(n)$, the proof for $SO(n)$ being entirely analogous. Firstly, note that $H^*(\Sigma BO(n+1)) \cong \Sigma\mathbf{Z}/2[w_1, \ldots, w_{n+1}]$, where w_i denotes the i^{th} Stiefel-Whitney class now in degree $i+1$, and by the Thom isomorphism, we have isomorphisms of graded rings

$$H^*(\Sigma MTO(n+1)) \cong \Sigma\mathbf{Z}[w_1, \ldots, w_{n+1}]\{\overline{U}\} \quad \text{and} \quad H^*(MTO(n)) \cong \mathbf{Z}/2[w_1, \ldots, w_n]\{\overline{U'}\}$$

where $\overline{U}$ and $\overline{U'}$ denote virtual Thom classes for $-\gamma^K_{n+1}$ and $-\gamma^K_n$ respectively, both in degrees $-n$. By comparing the connectivity of the various $\mathcal{A}$-modules, it is clear the map η^* is an isomorphism in degrees $k \leq 0$. Thus in particular, $\eta^*(w_i\overline{U}) = w_i\overline{U'}$ for all $1 \leq i \leq n$. As $\eta^* \colon H^*(\Sigma MTO(n+1)) \to H^*(MTO(n))$ is a ring homomorphism, and $H^*(MTO(n))$ is generated by the classes $w_i\overline{U'}$, it follows that η^* is surjective, hence $\partial^* = 0$ by exactness.

Alternatively, we can leverage the additional structure endowed by the Steenrod algebra. In particular, letting $\overline{w}_i := \mathrm{Th}^{-1}(Sq^i(\overline{U'}))$ denote the i^{th} Stiefel-Whitney class of the virtual vector bundle $-\gamma^K_n$, by definition $\overline{w}_i$ is given as the sum of degree i monomials in the total Stiefel-Whitney class

$$\overline{w} = w^{-1},$$

where the inverse of the total Stiefel-Whitney class w of the universal bundle γ^K_n is taken in $\mathbf{Z}/2[[w_1, w_1, \ldots]]$. As $\eta^*(\overline{U}) = \overline{U'}$, by exactness we have $\partial^*(\overline{U'}) = 0$. By $\mathcal{A}$-linearity, we have $\partial^*(\overline{w}_i\overline{U'}) = \partial^*(Sq^i(\overline{U'})) = Sq^i(\partial^*(\overline{U'})) = 0$ for each i. As the $\overline{w}_i$'s are defined inductively in terms of the universal Stiefel-Whitney classes of degree $\leq i$, this yields another proof that $\partial^*(w_i\overline{U'}) = 0$ for each i, hence $\partial^* = 0$. □

The upside of Proposition 4.1.3 is we can now leverage the geometric boundary theorem (2.3.4 in [Rav86]) in our spectral sequence computations. Namely, we have an algebraic boundary map

$$\delta \colon \operatorname{Ext}^{s,t}_{\mathcal{A}}(\Sigma BK(n+1)_+, \mathbf{Z}/2) \to \operatorname{Ext}^{s+1,t}_{\mathcal{A}}(MTK(n), \mathbf{Z}/2)$$

on E_2-pages of the respective Adams spectral sequences which commutes with differentials. Using the notation

$$E_2^{s,t} := \operatorname{Ext}^{s,t}_{\mathcal{A}}(H^*(\Sigma BK(n+1)), \mathbf{Z}/2), \tag{4.4}$$

$${}'E_2^{s,t} := \operatorname{Ext}^{s,t}_{\mathcal{A}}(H^*(\Sigma MTK(n+1)), \mathbf{Z}/2), \tag{4.5}$$

$${}''E_2^{s,t} := \operatorname{Ext}^{s,t}_{\mathcal{A}}(H^*(MTK(n)); \mathbf{Z}/2), \tag{4.6}$$

and $i := (\eta^*)^*$ and $j := (\beta^*)^*$ for the induced maps on Ext-groups via precomposition, the cofiber sequence above yields a long exact sequence of Ext-groups

$$\ldots \longrightarrow {}''E_2^{s,t} \xrightarrow{i} {}'E_2^{s,t} \xrightarrow{j} E_2^{s,t} \xrightarrow{\delta} {}''E_2^{s+1,t} \longrightarrow \ldots . \tag{4.7}$$

which is natural with respect to Adams differentials. In particular, the sequence persists to subsequent pages E_r, and taking $r \to \infty$ it converges to the long exact sequence of homotopy groups associated to the cofiber sequence. This will allow us, in favorable cases, to transfer differentials from Adams spectral sequences which have been computed in a range (such as for $\mathbf{S}^0$ or $MTSO(2)$) to those we are interested in computing (i.e., for $MTO(1)$ and $BSO(3)$).

4.2. **A warm-up computation: the homotopy groups of $MTO(1)$.** The Madsen-Tillman spectrum $MTO(1)$ is a 2-local spectrum which has been extensively studied in classical stable homotopy theory under the guise of the notation $\mathbf{RP}^\infty_{-1}$. Computing the homotopy groups of $MTO(1)$ is made significantly easier by the following result, well-known to experts.

Lemma 4.2.1. *The cofiber of the composite*

$$\widetilde{\beta} : MTO(1) \xrightarrow{\beta} \Sigma^\infty_+\mathbf{RP}^\infty \simeq \Sigma^\infty\mathbf{RP}^\infty \vee \mathbf{S}^0 \xrightarrow{\pi_2} \mathbf{S}^0$$

coincides with the cofiber C_{KP} of the Kahn-Priddy map $\Sigma^\infty \mathbf{RP}^\infty \to \mathbf{S}^0$ from [KP72]. *In particular, the composite induces an epimorphism on homotopy groups in positive degrees.*

Proof. In [Ada74], Adams showed that the Kahn-Priddy map is characterized, up to an equivalence of the source, by the fact that it induces an isomorphism on the first homotopy group $\pi_1(\Sigma^\infty \mathbf{RP}^\infty) \cong \mathbf{Z}/2 \cong \pi_1(\mathbf{S}^0)$. We claim the composite $\widetilde{\omega}\colon \Sigma^\infty \mathbf{RP}^\infty \xrightarrow{\mathrm{id}\vee 0} MTO(1) \xrightarrow{\omega_{O(1)}} \mathbf{S}^0$ satisfies this condition. Indeed, the MTW cofiber sequence associated to $MTO(1)$ yields the exact sequence

$$\pi_1(MTO(1)) \cong \mathbf{Z}/2 \xrightarrow{\beta_*} \pi_1(\Sigma^\infty_+ \mathbf{RP}^\infty) \cong (\mathbf{Z}/2)^2 \xrightarrow{\omega_{O(1)*}} \pi_1(\mathbf{S}^0) \cong \mathbf{Z}/2 \longrightarrow 0$$

hence it suffices to prove $\widetilde{\beta}$ induces an isomorphism on the first homotopy group. Looking at the Adams spectral sequence ${}'E_2^{s,t} = E_2^{s,t}(MTO(1))$ for $MTO(1)$ in figures 6 and 7, the long exact sequence of Ext-groups 4.7 yields the exact sequence

$$0 \longrightarrow {}'E_2^{1,3} \cong \mathbf{Z}/2\{x_{1,2}\} \xrightarrow{j} E_2^{1,3} \cong \mathbf{Z}/2$$

where $E_2^{1,3} = E_2^{1,3}(\Sigma \mathbf{RP}^\infty_+) \cong E_2^{1,3}(\Sigma \mathbf{RP}^\infty) \oplus E_2^{1,3}(\mathbf{S}^1)$ splits, with first summand being zero and the second summand generated by the class h_1 converging to the Hopf element $\eta \in \pi_2(\mathbf{S}^1)$. The class $x_{1,2}$ converges to the generator for $\pi_1(MTO(1))$, and j converges to β_* on π_1, as desired.

It then follows that we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] \arrow[d] & \Sigma^\infty \mathbf{RP}^\infty \arrow[r, equal] \arrow[d, "\mathrm{id}\vee 0"'] & \Sigma^\infty \mathbf{RP}^\infty \arrow[d, "\widetilde{\omega}"] \\
MTO(1) \arrow[r, "\beta"] \arrow[d, equal] & \Sigma^\infty \mathbf{RP}^\infty \vee \mathbf{S}^0 \arrow[r, "\omega_{O(1)}"] \arrow[d, "\pi_2"'] & \mathbf{S}^0 \arrow[d] \\
MTO(1) \arrow[r, "\widetilde{\beta}"'] & \mathbf{S}^0 \arrow[r, dashed] & C_{\mathrm{KP}}
\end{tikzcd} \tag{4.2}$$

where the columns and first two rows are cofiber sequences, and hence by the (3×3)-lemma, there exists a dashed arrow making the bottom row is a cofiber sequence, completing the proof. ∎

The above lemma gives us a lower bound on the homotopy groups of $MTO(1)$. Indeed, by proving $\widetilde{\omega}$ is a Kahn-Priddy map, the MTW cofiber sequence for $MTO(1)$ implies that β_* surjects onto $\pi_*(\mathbf{S}^0) \oplus \ker(\omega_*)$ in positive degrees. Even more importantly, it implies that the map $j\colon {}'E_2 \to E_2 \cong E_2(\Sigma \mathbf{RP}^\infty) \oplus E_2(\mathbf{S}^1)$ of E_2-pages converging to β_* on homotopy groups, must surject onto all classes in $E_2(\mathbf{S}^0)$ in positive degrees which persist to the E_∞-page. This makes the geometric boundary theorem a very powerful tool to lift differentials from the Adams spectral sequence of S^0 to ${}'E_2$. We combine this information, together with h_i-linearity of Adams differentials, to determine differentials in the Adams spectral sequence for $MTO(1)$ up to topological degree 38.

Notation 4.2.3. The notation for classes in the Adams spectral sequence for $\mathbf{S}^0$ and the corresponding elements in $\pi_*(\mathbf{S}^0)$ are taken from page 82, table IV of [Whi70].

Proposition 4.2.4. *Keeping in-line with the notation 4.4, let ${}'E_2^{s,t} = E_2^{s,t}(\Sigma MTO(1))$, with Adams chart depicted in figures 6 and 7 in Appendix A.*

1. *There are no nontrivial differentials supported in degrees $t-s \le 15$.*
2. *The only non-trivial differentials supported in degrees $16 \le t-s \le 17$ is a d_2-differential on ${}'E_2^{1,17}$ and ${}'E_2^{4,22}$, and two d_3-differentials, on ${}'E_2^{2,18}$ and ${}'E_2^{3,19}$.*
3. *There are precisely three nontrivial differentials supported in degrees $18 \le t-s \le 22$. They are d_2-differentials, supported on ${}'E_2^{4,22}$, ${}'E_2^{7,31}$, and ${}'E_2^{8,32}$.*
4. *In topological degrees $23 \le t-s \le 30$, the only nontrivial differentials are d_2's, supported on the unique classes in ${}'E_2^{8,32}$, ${}'E_2^{9,33}$, ${}'E_2^{8,34}$, ${}'E_2^{7,34}$, ${}'E_2^{8,36}$, ${}'E_2^{9,37}$, $E_2^{7,37}$, and $E_2^{8,38}$.*

Proof.

1. There are several candidate differentials in degrees $t-s \le 15$ whose vanishing deserves some justification. Firstly, by surjectivity of β_* onto $\pi_3(\mathbf{S}^0) \cong \mathbf{Z}/8$, it follows that ${}'E_2^{3,7}$ cannot be the target of a nonzero differential, hence $d_2^{5,1} = 0$. The remaining possible nonzero differentials are supported

in degrees $9 \leq t-s \leq 11$. There are no generators in degree $t-s=8$ which are the target of a differential. Indeed, lower bound on $\pi_7(MTO(1))$ given by the exact sequence

$$\pi_7(MTO(1)) \longrightarrow (\mathbf{Z}/16)^2 \oplus \mathbf{Z}/2 \longrightarrow \mathbf{Z}/16$$

together with the Adams chart for $\Sigma MTO(1)$ imply $\pi_7(MTO(1)) \cong \mathbf{Z}/16 \oplus \mathbf{Z}/2$, from which the above result follows. Finally, we can verify that there are no differentials supported in degrees $t-s=10$ and $t-s=11$ by comparing the bounds on the size of $\pi_9(MTO(1))$ and $\pi_{10}(MTO(1))$ given by the $'E_2$-page and the exact sequence of homotopy groups associated to the cofiber sequence $MTO(1) \to \mathbf{RP}^\infty \to S^0$. Via the latter, we have an exact sequence

$$\mathbf{Z}/8 \xrightarrow{0} \pi_{10}(MTO(1)) \longrightarrow (\mathbf{Z}/2)^2 \oplus \mathbf{Z}/8 \twoheadrightarrow \mathbf{Z}/2 \xrightarrow{0} \pi_9(MTO(2)) \longrightarrow (\mathbf{Z}/2)^7 \twoheadrightarrow (\mathbf{Z}/2)^3 \ .$$

The claim follows by observing the $'E_2$-page has 4 generators each in degrees $t-s=10$ and $t-s=11$ and hence neither column can support a nontrivial differential.

(2) Lemma 4.2.1 implies we have a short exact sequence

$$0 \longrightarrow \pi_{14}(MTO(1)) \longrightarrow (\mathbf{Z}/2)^4 \twoheadrightarrow (\mathbf{Z}/2)^2 \longrightarrow 0 \ .$$

We conclude $\pi_{14}(MTO(2)) \cong (\mathbf{Z}/2)^2$. It follows that there exists precisely three nontrivial differentials with target in degree $t-s=15$. We leverage the geometric boundary theorem: the long exact sequence of Ext-groups induce an exact sequence

$$0 \xrightarrow{i} {'E}_2^{1,17} \cong \mathbf{Z}/2\{x_{1,17}\} \xrightarrow{j} E_{2,\mathbf{S}^0}^{1,17} \oplus {''E}_{2,\mathbf{RP}^\infty}^{2,19} \cong \mathbf{Z}/2\{\widehat{h}_4\} \oplus \mathbf{Z}/2\{h_4\}$$

and so $j(x_{1,7}) = h_4$ by Lemma 4.2.1. It follows that the d_2 differential supported on h_4 lifts to ${'E}_2^{1,17}$. By h_i-linearity, the remaining differentials are d_3's, one of which is supported on the unique class $x_{2,18} \in {'E}_2^{2,18}$ with a nontrivial h_0, the other supported on $h_0 x_{2,18}$.

(3) We have the following exact sequence of homotopy groups

$$\pi_{20}(MTO(2)) \longrightarrow \mathbf{Z}/2 \oplus (\mathbf{Z}/8)^2 \longrightarrow \mathbf{Z}/8 \ .$$

Compared with the number of generators in degree $t-s=21$, it follows that there cannot exist a nonzero differential with source or target in this topological degree. An analogous argument implies the same holds for topological degree $t-s=20$. Moving leftwards, the map of spectral sequences $j\colon E_2^{s,t} \to {''E}_2^{s,t}$ is injective in bidegree $(4,23)$. We have $E_2^{s,t} = \mathbf{Z}/2\{x,y\}$, where y denotes the unique class with a nontrivial h_0-action. As $h_0 y$ is h_1-divisible, by h_i-linearity, the same must be true for $h_0 i(y)$, hence it follows that $i(y) = f_0$. Since f_0 supports a nonzero d_2-differential, by naturality of i, this lifts to a nonzero d_2-differential $d_2\colon E_2^{4,23} \to E_2^{7,22}$. Multiplying by h_0, we get another nonzero d_2-differential supported on $E_2^{5,23}$. The exact sequence

$$\mathbf{Z}/2 \oplus \mathbf{Z}/8 \xrightarrow{0} \pi_{18}(MTO(2)) \longrightarrow (\mathbf{Z}/2)^2 \oplus (\mathbf{Z}/8)^3 \longrightarrow \mathbf{Z}/2 \oplus \mathbf{Z}/8$$

implies these are the only nonzero differentials supported in degree $t-s=19$. The class $x_{4,22}$ generating ${'E}_2^{4,22}$ admits nontrivial h_0-action. As $\pi_{17} MTO(1)$ injects into $\pi_{17}(\Sigma_+^\infty \mathbf{RP}^\infty) \cong (\mathbf{Z}/2)^{10}$, it follows that either $x_{4,22}$ or $h_0 x_{4,22}$ cannot persist to the E_∞-page. This forces ${'E}_2^{4,22}$ to support a nontrivial d_2-differential. Comparing the number of classes in degree $t-s=17$ with the lower bound on the size of $\pi_{16}(MTO(1))$ implied by Lemma 4.2.1 implies there are no other differentials in degrees $t-s=18$ and 17.

(4) We only prove the existence of the nontrivial differentials, as the task of verifying this list is exhaustive is a standard application of leveraging the short exact sequences of homotopy groups given by the MTW cofiber sequence and Lemma 4.2.1. The map $j\colon {'E}_2^{s,t} \to {''E}_2^{s,t}$ is injective in bidegrees $(7,31)$ and $(7,26)$, and so the differentials supported in the summand $E_{2,\mathbf{S}^0}^{s,t}$ in these bidegrees lift along j. Linearity with respect to h_0 and h_2 implies the existence of nontrivial d_2-differentials supported in bidegrees $(8,32), (8,35), (9,36)$, and $(8,38)$. The existence of the remaining two nontrivial differentials supported on ${'E}_2^{8,34}$ and ${'E}_2^{7,37}$ follows by comparing the number of classes in topological degrees 26 and 31 in ${'E}_2^{s,t}$ with the lower bounds on $\pi_{25} MTO(1)$ and $\pi_{30} MTO(1)$ implied by Lemma 4.2.1.

□

4.3. **The homotopy groups of $MTO(2)$.** We are now ready to compute the homotopy groups of $MTO(2)$ via the splitting into $D(2)$ and $\Sigma^\infty_+ BSO(3)$. We begin by determining differentials in the Adams spectral sequence for $D(2)$. In topological degrees $t-s \leq 18$, there is only room for a single differential with nonzero target in the form of a d_2-differential on $E_2^{14,1}$.

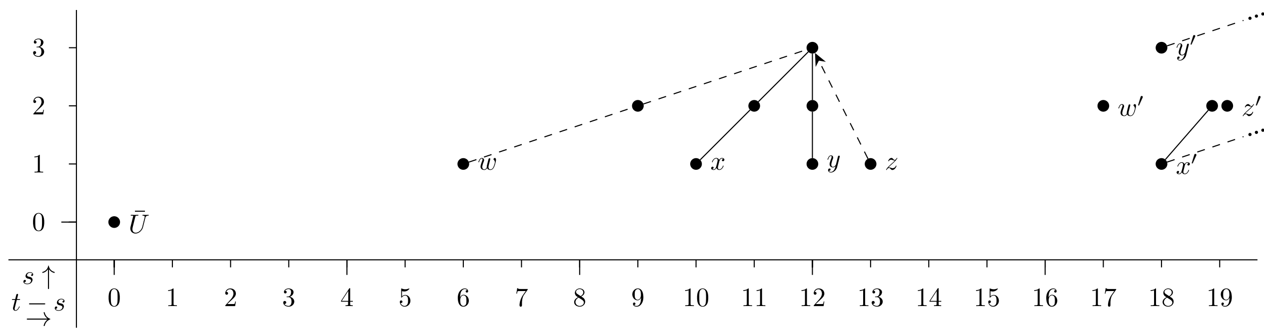

FIGURE 1. $\mathrm{Ext}_{\mathcal{A}}(D(2), \mathbf{Z}/2) = \mathrm{Ext}_{\mathcal{A}}(D_2, \mathbf{Z}/2)$

Proposition 4.3.1. *The d_2-differential $d_2\colon E_2^{1,14}(D(2)) \to E_2^{3,15}(D(2))$ vanishes.*

Proof. The cofiber sequence $D(1) \to D(2) \to \Sigma^2 M(2)$ yields a long exact sequence of Ext-groups relating the E_2-pages of the respective Adams spectral sequences of $D(1)$, $D(2)$, and $\Sigma^2 M(2)$. We also have an associated long exact sequence of homotopy groups

$$\ldots \longrightarrow \pi_{13}(D(1)) \longrightarrow \pi_{13}(D(2)) \longrightarrow \pi_{13}(\Sigma^2 M(2)) \longrightarrow \pi_{12}(D(1)) \longrightarrow \ldots .$$

As $D(1) \simeq \Sigma MTO(1)$, we know the homotopy groups of $D(1)$ in the relevant ranges, namely we have the following exact of homotopy groups

$$0 \longrightarrow \pi_{13}(D(2)) \longrightarrow \pi_{13}(\Sigma^2 M(2)) \longrightarrow \mathbf{Z}/8 \longrightarrow \pi_{12}(D(2)) \longrightarrow \pi_{12}(\Sigma^2 M(2)) \longrightarrow \mathbf{Z}/2 \oplus \mathbf{Z}/8 \longrightarrow \mathbf{Z}/2 \ .$$

It suffices to conclude $\pi_{13}(\Sigma^2 M(2)) = \mathbf{Z}/2 \ltimes \mathbf{Z}/8$. Looking at the Adams spectral sequence for $\Sigma^2 M(2)$ A, the splitting $M(2) \simeq L(2) \vee \mathbf{RP}^\infty$ induces a splitting of E_2-pages

$$\mathrm{Ext}^{s,t}_{\mathcal{A}}(H^*(\Sigma^2 M(2)), \mathbf{Z}/2) \cong \mathrm{Ext}^{s,t}_{\mathcal{A}}(H^*(\Sigma^2 L(2)), \mathbf{Z}/2) \oplus \mathrm{Ext}^{s,t}_{\mathcal{A}}(H^*(\Sigma^2 \mathbf{RP}^\infty), \mathbf{Z}/2).$$

This splitting persists on every page E_r of the spectral sequence, such that for $r = \infty$ it corresponds (up to Adams filtration) to the splitting of homotopy groups. The Adams spectral sequence and stable homotopy groups of $\mathbf{RP}^\infty$ are well-known (page 82, table IV of [Whi70]). In particular, $E_r^{*,*}(\mathbf{RP}^\infty)$ admits no nontrivial differentials in topological degrees $t - s \leq 13$, and from this we can conclude that there is no differential supported in topological degree 13 in the Adams spectral sequence for $\Sigma^2 M(2)$. This completes the proof. $\square$

Our approach relies heavily on the work of Rognes in [Rog02] (pages 894-898), in which he computed the Adams spectral sequence for $MTSO(2)$ and determined all differentials up through degree 20.

The fact that differentials are linear with respect to the $E_2^{*,*}(\mathbf{S}^0)$-module structure of $E_2^{*,*}$ precludes the existence of nontrivial differentials in topological degrees $0 \leq t-s \leq 11$ and $13 \leq t-s \leq 15$. The following proposition determines all nontrivial differentials in the Adams spectral sequence for $BSO(3)$ up to and including topological degree 16.

Proposition 4.3.2.

(1) There exists a nontrivial d_2-differential $d_2\colon E_2^{1,14} \to E_2^{3,15}$ with target $h_0 h$. Consequently by h_0-linearity, there are nontrivial d_2-differentials supported on $E_2^{1,15}$ and $E_2^{2,16}$.

(2) There are no nontrivial differentials supported on $E_2^{1,18}$.

Proof.

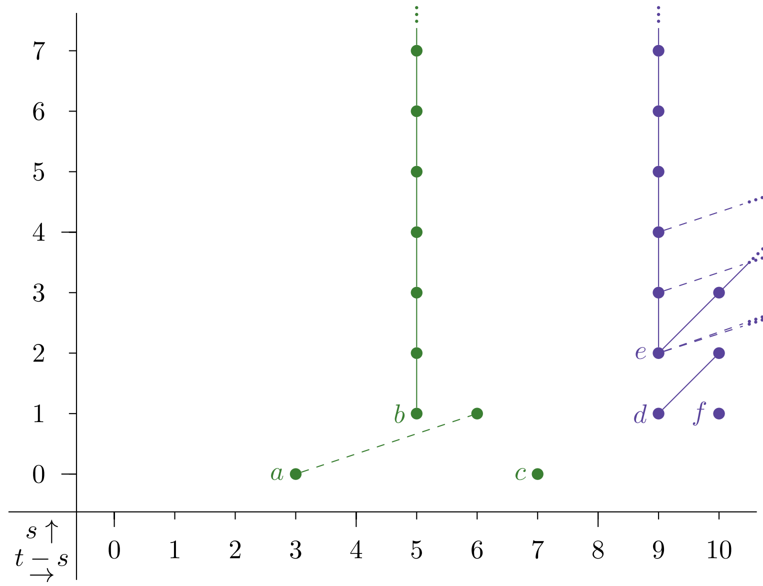

FIGURE 2.1. $E_2 = \mathrm{Ext}_{\mathcal{A}}(H^*(\Sigma BSO(3)), \mathbf{Z}/2)$

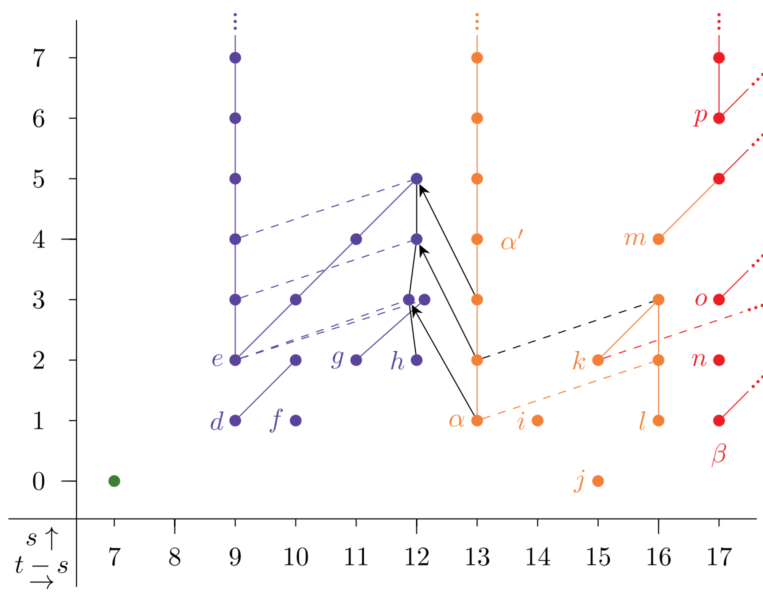

FIGURE 2.2. $E_2 = \mathrm{Ext}_{\mathcal{A}}(H^*(\Sigma BSO(3)), \mathbf{Z}/2)$

(1) The map of spectral sequences $\delta\colon E_2^{s,t} \to {}''E_2^{s+1,t}$ is an isomorphism in bidegree $(1,14)$. Indeed, the long exact sequence 4.7 of Ext-groups yields the fiber sequence

$$ {}'E_2^{1,14} = 0 \longrightarrow E_2^{1,14} \cong \mathbf{Z}/2\{\alpha\} \xrightarrow{\delta} {}''E_2^{2,14} \cong \mathbf{Z}/2\{h_{06}h_0\} \ .$$

As the image of the generator of $E_2^{1,14} \cong \mathbf{Z}/2\{\alpha\}$ along δ supports a nonzero d_2-differential on $MTSO(2)$, this implies α does as well. It remains to determine the image of this differential. We have $E_2^{2,14} \cong \mathbf{Z}/2\{h\}$ and ${}''E_2^{3,14} \cong \mathbf{Z}/2\{h_{02}h_0h_3\} \oplus \mathbf{Z}/2i\{h_1^3\}$ using Rognes' notation. The map $\delta\colon E_2^{2,14} \to {}''E_2^{3,14}$ is an injection by the same argument above (the fiber is a subgroup of ${}'E_2^{2,14} = 0$). As $h_1h = 0$, it follows that $i(h)$ must have trivial h_1-action, and so we conclude $i(h) = h_{02}h_0h_3$. Then h_0-linearity of i implies $i(h_0h) = h_{02}^2h_0h_3 = {}''d_2(i(\alpha))$, hence $d_2(\alpha) = h_0h$ as required.

(2) We claim the map $\delta\colon E_2^{s,t} \to {}''E_2^{s+1,t}$ is an isomorphism in bidegrees (3,19) and (4,20) and injective in bidegree (1,18). Let α denote the generator of $E_2^{1,18}$.

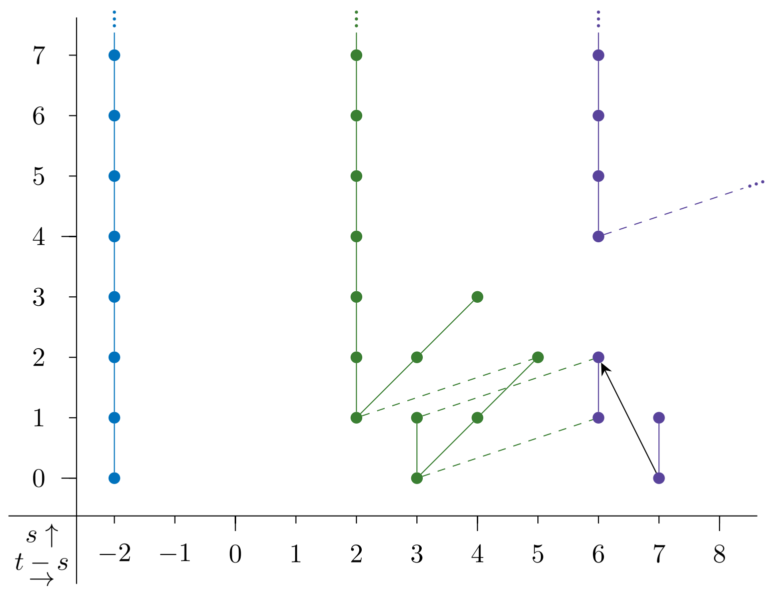

FIGURE 3.1. $'E_2 = \mathrm{Ext}_{\mathcal{A}}(H^*(\Sigma MTSO(3)), \mathbf{Z}/2)$

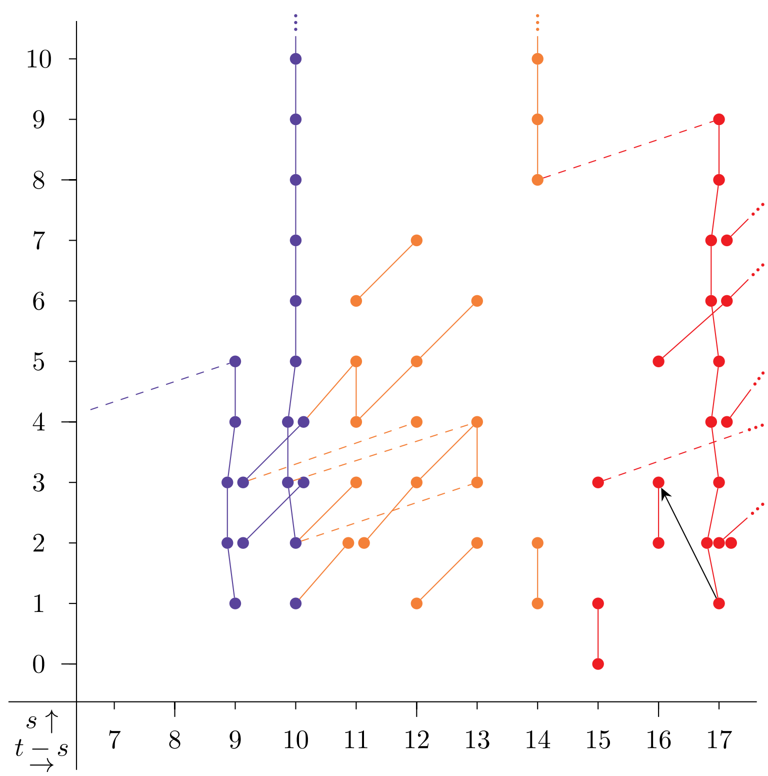

FIGURE 3.2. $'E_2 = \mathrm{Ext}_{\mathcal{A}}(H^*(\Sigma MTSO(3)), \mathbf{Z}/2)$

Since the d_2-differential on $''E_2^{2,18}$ is zero, by injectivity of δ in degrees $(1,17)$ and $(3,18)$, it follows that $d_2\colon E_2^{1,18} \to E_2^{3,19}$ is zero as well.

We have $E_3^{4,20} \cong E_2^{4,20}$ and $''E_3^{5,20} \cong {''E_2^{5,20}}$, hence δ descends to an isomorphism between these terms on the E_3-pages. Furthermore, we have $E_3^{1,18} \cong E_2^{1,18}$, $''E_3^{2,18} \cong {''E_2^{2,18}}$, and so $\delta\colon E_3^{1,18} \to E_3^{2,18}$ remains an injection. Since $0 = d_3(\delta(\alpha)) = \delta(d_3(\alpha))$, we conclude $d_3(\alpha) = 0$. This completes the proof.

$\square$

To determine the homotopy groups of $D(2)$ and $BSO(3)$ from the Adams E_∞-pages we determined in the range $t-s \leq 16$, we need to solve the corresponding extension problems to pass from the associated graded objects of the Adams filtration to the actual homotopy groups. Again, here the $E_r^{*,*}(\mathbf{S}^0)$-module structure of Adams spectral sequences comes in handy. As the h_0-action on E_2-pages converges to multiplication by 2 on homotopy groups induced by the natural $\mathbf{S}^0$-module structure of spectra, we can rule out the existence of certain hidden extensions. For example, if there existed a hidden h_0-extension $e = h_0 d$ in degree $t-s=9$ of the Adams spectral sequence for $\Sigma BSO(3)$, since $h_1 e \neq 0$, this would imply at the level of homotopy groups the relation $2\eta d \neq 0$ in $\pi_9 BSO(3)$, but the Hopf map $\eta \in \pi_1(\mathbf{S}^0)$ has order 2, hence such an extension cannot not exist.

With this in mind, the following theorem summarizes our computations of the 2-primary homotopy groups of $MTO(2)$ through the 16^{th} stem. Extension problems which we could not resolve are distinguished by the symbol '$\ltimes$'. Relative to the 2-complete splitting $MTO(2) \simeq \Sigma^{-2}D(2) \vee \Sigma^\infty BSO(3) \vee \mathbf{S}^0$, generators of $\pi_*(MTO(2))$ which lie in $\pi_*(\Sigma^{-2}D(2))$, $\pi_*(\mathbf{S}^0)$, or $\pi_*(\Sigma^\infty BSO(3))$, are denoted blue, green and orange respectively.

Theorem 4.3.3. *The 2-primary homotopy groups $\pi_*(MTO(2))$ in dimensions $* \leq 15$ are given as follows:*

(4.4)

k	$\pi_k MTO(2)_{(2)}$
-2	$\mathbf{Z}/2\{U\}$
-1	0
0	$\mathbf{Z}\{\iota\}$
1	$\mathbf{Z}/2\{\eta\}$
2	$\mathbf{Z}/2\{a\} \oplus \mathbf{Z}/2\{\eta^2\}$
3	$\mathbf{Z}/8\{\nu\}$
4	$\mathbf{Z}/2\{w\} \oplus \mathbf{Z}\{b\}$
5	$\mathbf{Z}/2\{\nu a\}$
6	$\mathbf{Z}/2\{c\} \oplus \mathbf{Z}/2\{\nu^2\}$
7	$\mathbf{Z}/2\{\nu w\} \oplus \mathbf{Z}/16\{\sigma\}$
8	$\mathbf{Z}/2\{x\} \oplus \mathbf{Z}/2\{d\} \oplus \mathbf{Z}\{e\} \oplus \mathbf{Z}/2\{\eta\sigma\} \oplus \mathbf{Z}/2\{\epsilon\}$
9	$\mathbf{Z}/2\{\eta x\} \oplus \mathbf{Z}/2\{f\} \ltimes \mathbf{Z}/2\{\eta d\} \oplus \mathbf{Z}/2\{\eta e\} \oplus \mathbf{Z}/2\{\nu^3\} \oplus \mathbf{Z}/2\{\eta\epsilon\} \oplus \mathbf{Z}/2\{\mu\}$
10	$\mathbf{Z}/8\{y\} \oplus \mathbf{Z}/2\{g\} \ltimes \mathbf{Z}/2\{\eta^2 e\} \oplus \mathbf{Z}/16\{\nu\mu\}$
11	$\mathbf{Z}/2\{z\} \oplus \mathbf{Z}/2\{h\} \ltimes \mathbf{Z}/2\{\eta g\} \oplus \mathbf{Z}/8\{\zeta\}$
12	$\mathbf{Z}\{\alpha'\}$
13	$\mathbf{Z}/2\{i\}$
14	$\mathbf{Z}/2\{j\} \oplus \mathbf{Z}/2\{k\} \oplus \mathbf{Z}/16\{\sigma^2\} \oplus \mathbf{Z}/16\{\kappa\}$
15	$\mathbf{Z}/2\{w'\} \oplus \mathbf{Z}/8\{l\} \oplus \mathbf{Z}/2\{m\} \oplus \mathbf{Z}/32\{\rho\} \oplus \mathbf{Z}/2\{\eta\kappa\}$

.

5. Looking ahead: the homology of the infinite non-orientable mapping class group

In Theorem C of [MP84], Mitchell and Priddy obtain a 2-complete splitting of $\Sigma_+^\infty BO(2)$:

$$\Sigma_+^\infty BO(2) \simeq \Sigma_+^\infty BSO(3) \vee M(2).$$

As finite products and coproducts in a triangulated category coincide, passing to infinite loop spaces (and noting that $\pi_0(M(2)) = 0$), this gives a homotopy equivalence

$$Q_0(BO(2)_+) \simeq Q_0(BSO(3)_+) \times \Omega^\infty M(2).$$

Lemma 4.10 of [CLM76] defines the $\mathbf{Z}/2$-homology of the (basepoint component) of the free infinite loop space $Q_0 X$ functorially in terms of $H_*(X;\mathbf{Z}/2)$. Thus the above equivalence, combined with the Künneth isomorphism, allows us to determine the rank of the homology groups $H_n(\Omega^\infty M(2))$. From this, we should in principal be able to use the fibration sequence

$$0 \longrightarrow \Omega_0^{\infty+1} MTO(1) \longrightarrow \Omega_0^{\infty+2}(D(2)) \longrightarrow \Omega_0^\infty(M(2)) \longrightarrow 0$$

associated to the cofiber sequence in 3.7 to compute the rank of the homology groups of $\Omega_0^\infty(D(2))$. The 2-complete splitting $MTO(2) \simeq \Sigma_+^\infty BO(2)_+ \vee \Sigma^{-2}D(2)$ descends to the following homotopy equivalence of infinite loop spaces:

$$\Omega_0^\infty MTO(2) \simeq Q_0(BO(2)) \times \Omega_0^{\infty+2}(D(2)),.$$

The homotopy equivalence $B\mathcal{N}^+ \simeq \Omega_0^\infty MTO(2)$ implies the above outlines a potential method for computing the rank of the $\mathbf{Z}/2$-homology groups of $B\mathcal{N}^+$ (and hence $B\mathcal{N}$). Since away from the prime 2, we have $B\mathcal{N}^+\left[\frac{1}{2}\right] \simeq Q_0(\mathbf{HP}^\infty)$, this would resolve one of the largest remaining obstructions in determining the integral homology of the infinite non-orientable mapping class group (up to potential non-trivial extensions of the 2-primary part).

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Appendix A. Adams charts

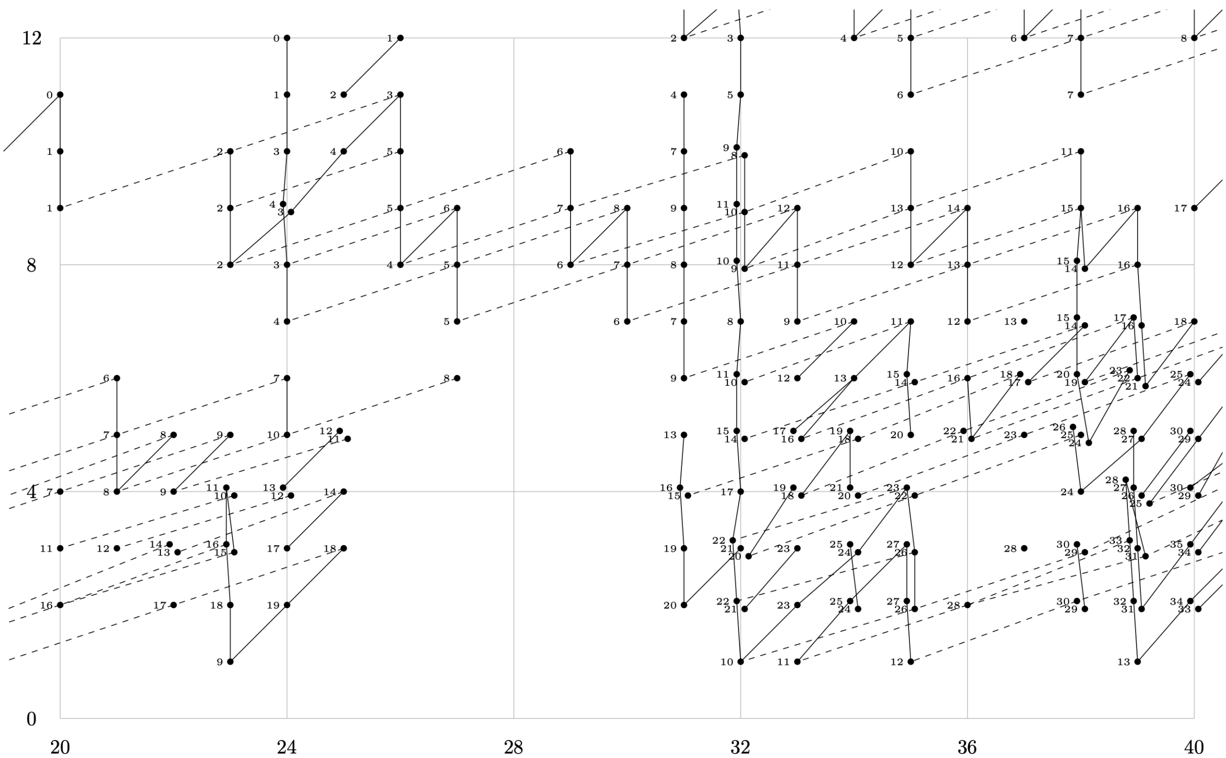

Figure 6. $\mathrm{Ext}_{\mathcal{A}}(H^*(\Sigma MTO(1), \mathbf{Z}/2), \mathbf{Z}/2)$ in degrees $0 \leq t - s \leq 20$.

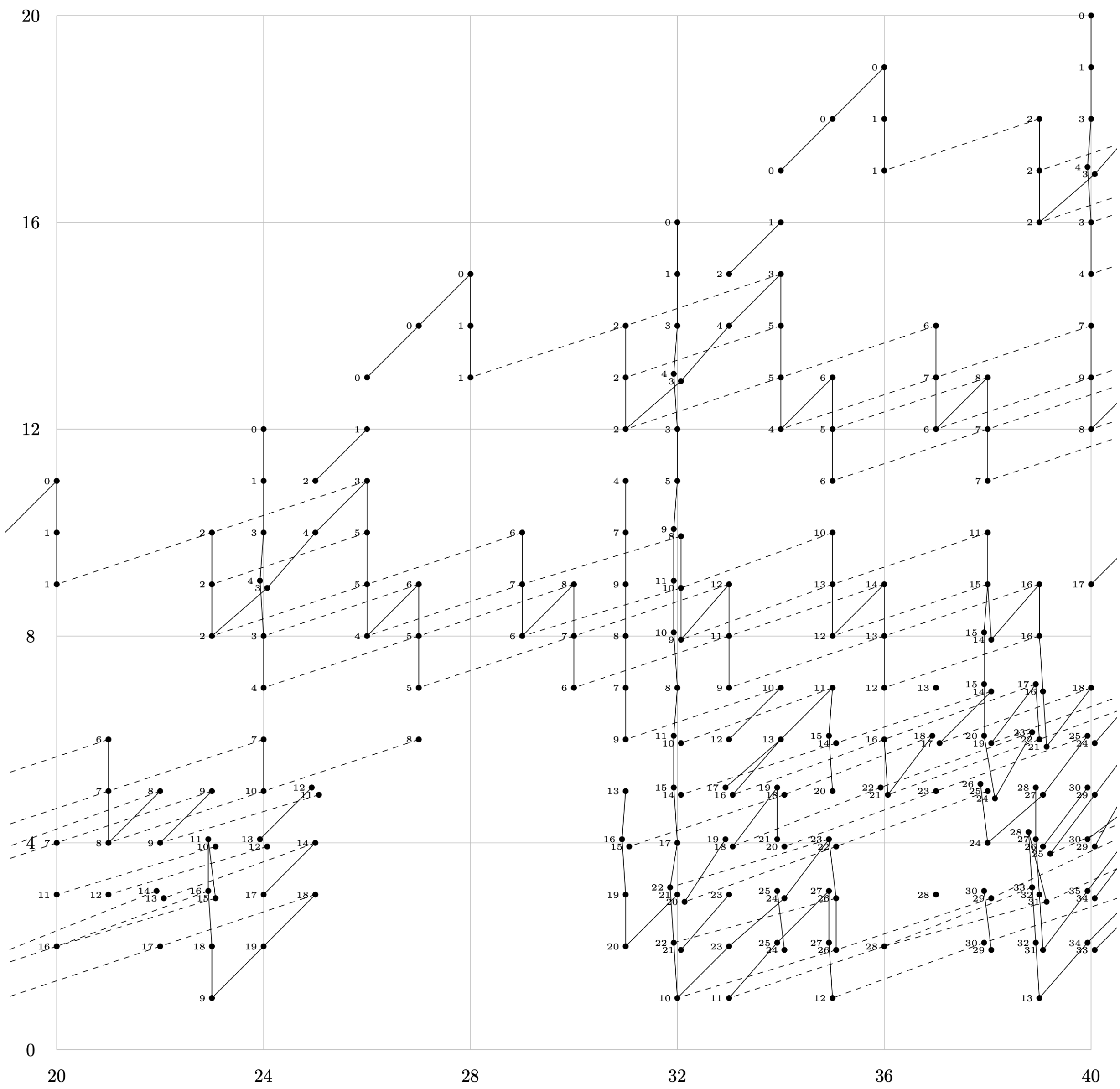

FIGURE 7. $\mathrm{Ext}_{\mathcal{A}}(H^*(\Sigma MTO(1), \mathbf{Z}/2), \mathbf{Z}/2)$ in degrees $20 \le t-s \le 40$.

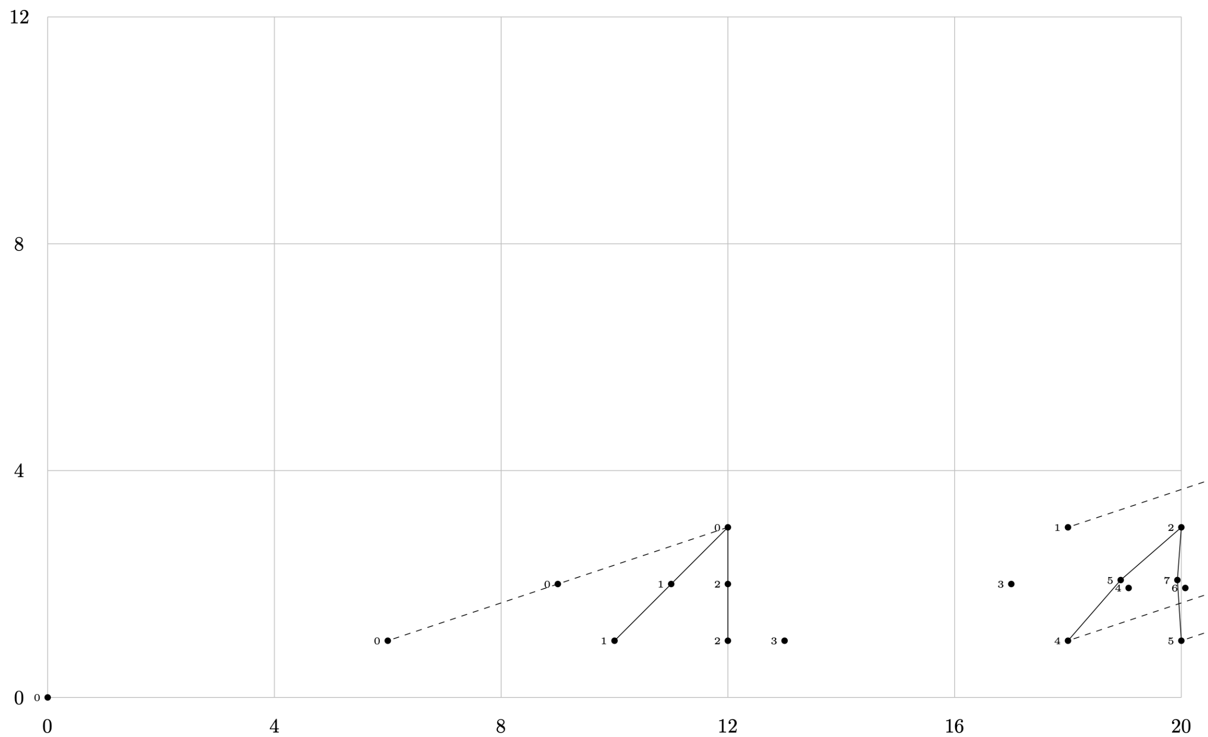

FIGURE 8. $\mathrm{Ext}_{\mathcal{A}}(H^*(D(2), \mathbf{Z}/2), \mathbf{Z}/2)$ in degrees $0 \leq t - s \leq 20$.

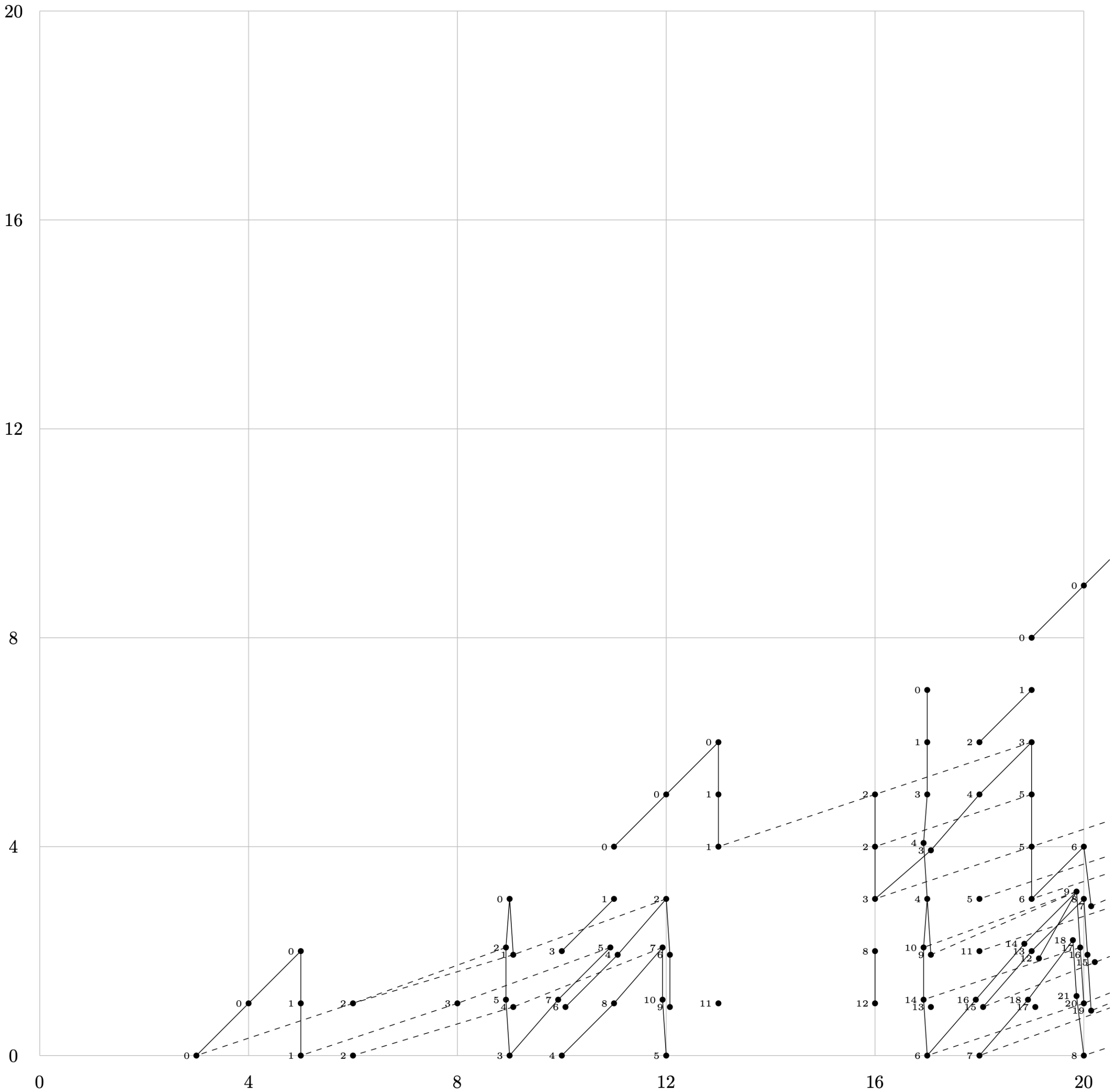

FIGURE 9. $\mathrm{Ext}_{\mathcal{A}}(H^*(\Sigma^2 M(2), \mathbf{Z}/2), \mathbf{Z}/2)$ in degrees $0 \leq t-s \leq 20$.

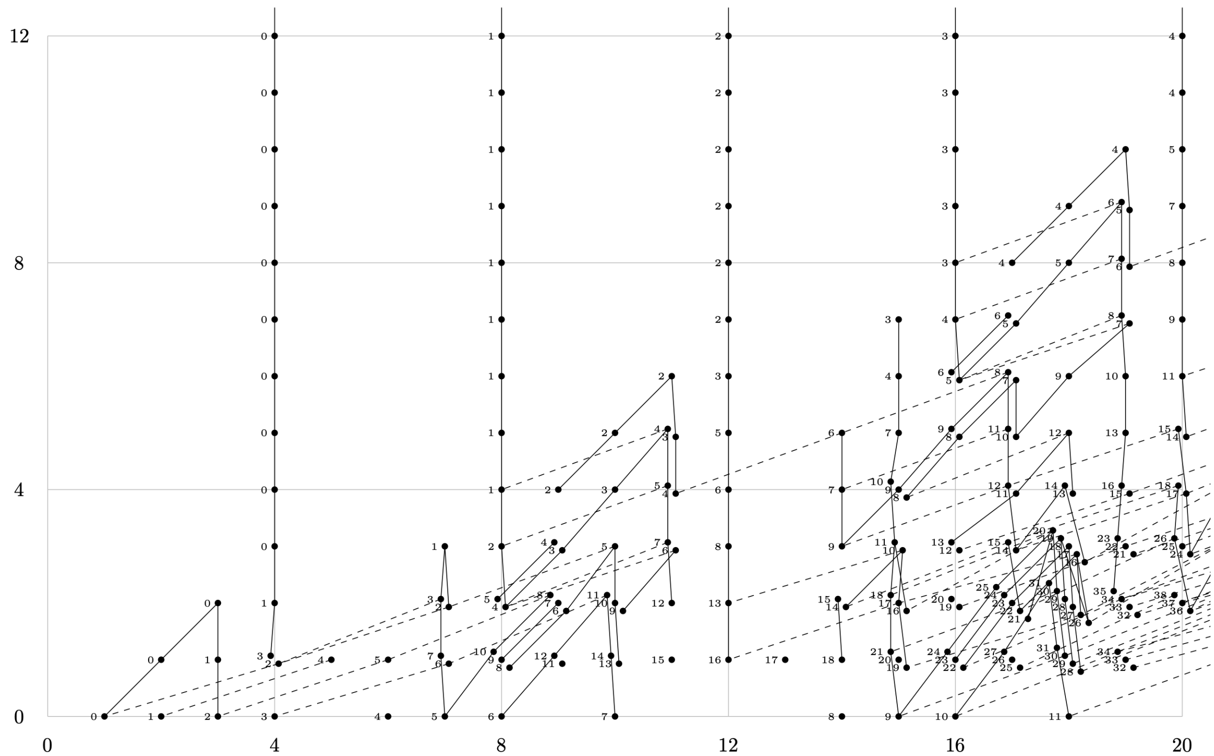

FIGURE 10. $\mathrm{Ext}_{\mathcal{A}}(\widetilde{H}^*(BO(2), \mathbf{Z}/2), \mathbf{Z}/2)$ in degrees $0 \le t-s \le 20$.

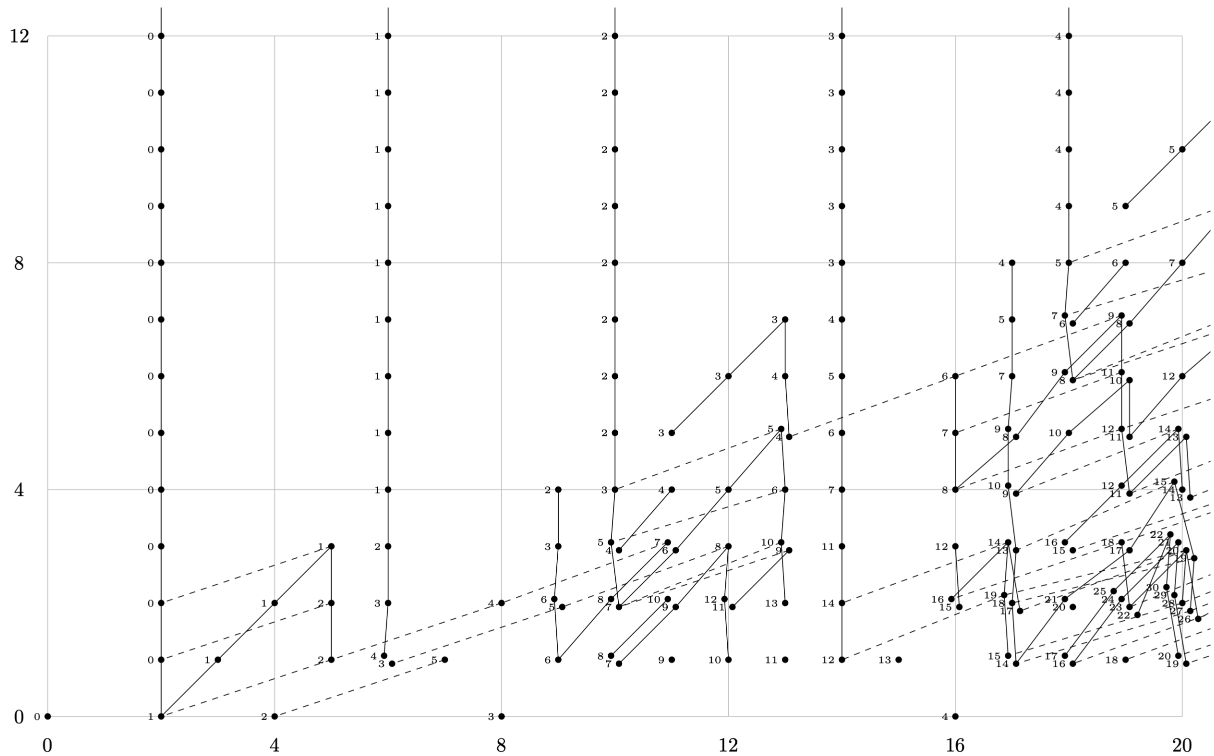

FIGURE 11. $\mathrm{Ext}_{\mathcal{A}}(H^*(\Sigma^2 MTO(2), \mathbf{Z}/2), \mathbf{Z}/2)$ in degrees $0 \leq t-s \leq 20$.

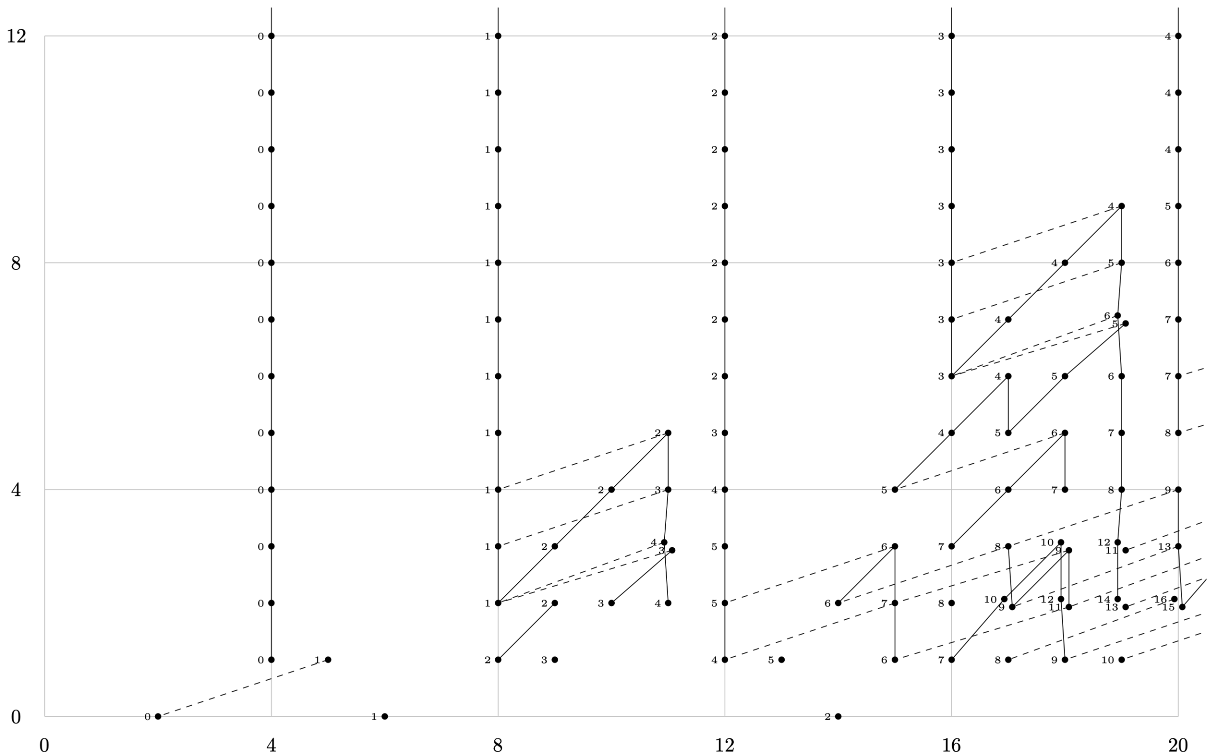

FIGURE 12. $\mathrm{Ext}_{\mathcal{A}}(\widetilde{H}^*(BSO(3), \mathbf{Z}/2), \mathbf{Z}/2)$ in degrees $0 \leq t - s \leq 20$.

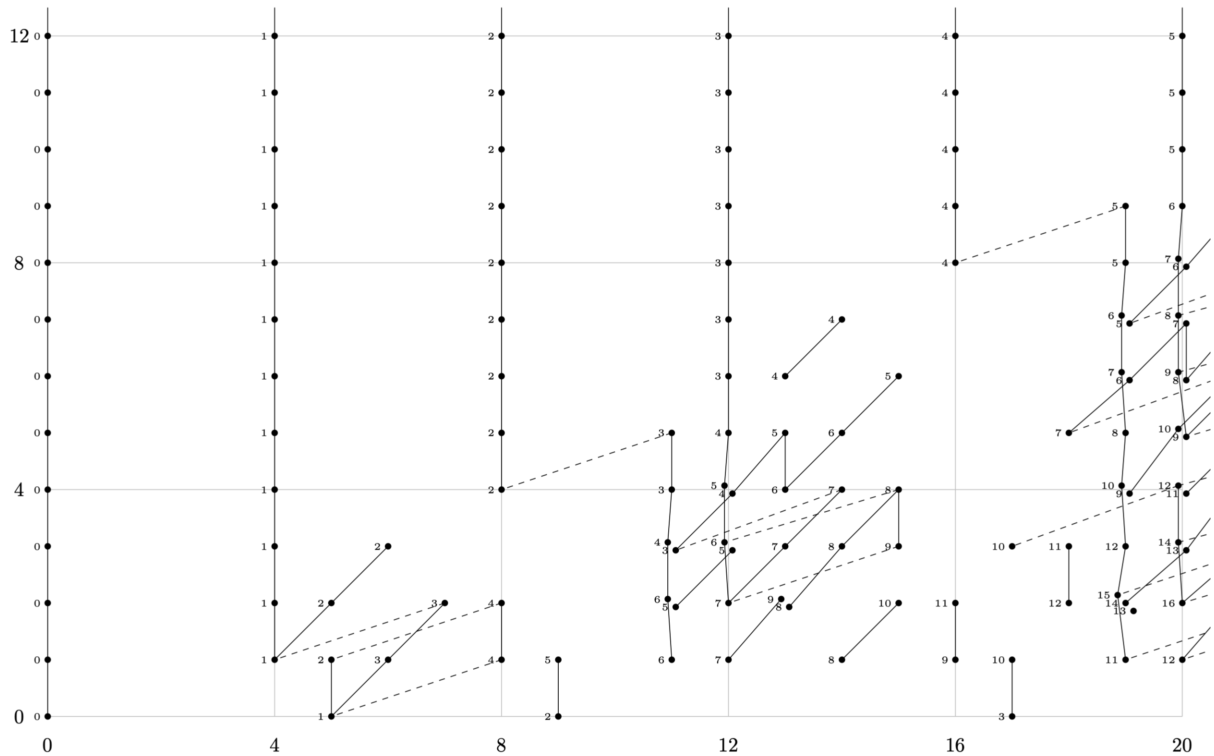

FIGURE 13. $\mathrm{Ext}_{\mathcal{A}}(H^*(\Sigma^3 MTSO(3), \mathbf{Z}/2), \mathbf{Z}/2)$ in degrees $0 \leq t - s \leq 20$.